Coevolving Human–LLM Organizational Knowledge Copilot: Certainty Collapsing Boundary Expansion

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1 Executive Summary

Despite wide Generative AI adoption, over 90% of high-value use cases remain stuck in the pilot stage [1]. The primary barrier is unreliability, as Large Language Model (LLM) hallucinations are an unacceptable liability in core business applications.

This write-up details a novel graph-based architecture designed to solve this trust gap. Instead of relying on the LLM's unreliable internal knowledge, our system first deconstructs multiple LLM answers into individual factual claims. It then clusters these claims to map the model's "beliefs", revealing a clear topology of uncertainty. This allows us to identify claims in different categories, e.g., "in agreement and factual" (high consensus, verified), "in agreement but hallucinated" (a consistent, shared error), "contested" claims (disagreeing and hallucinated, disagreeing but factual, novel, ambiguous, or low-consensus), and many other nuanced states.

Our framework (refer to Figure 1 for overview) resolves this uncertainty through a process called **"collapsing"**; converting ambiguity into a clear determination (true, false, or to be human verified). This is achieved by grounding the claims against **"knowledge priors"**: trusted external knowledge such as compliance documents, Standard Operating Procedures (SOPs), archival data, procedural automation workflows, and expert-verified records. The system cascades this process, automatically validating claims that align with priors and collapsing those that contradict them. This process balances autonomy and safety, maximizing automation for known information while routing only the most informative claims to human experts. This human-in-the-loop design, optimized by active learning, minimizes the cost of human effort.

The result is a human-AI hybrid system that builds an evolving map of organizational knowledge, turning expert-validated information into new priors that improve reliability over time. This framework is applicable to high-value workflows, enabling dynamic compliance monitoring, identifying new automation targets, and expanding internal knowledge. Our proposed method moves beyond simple chatbots, providing a pathway to build reliable, human-centric copilots capable of being trusted with critical operations.

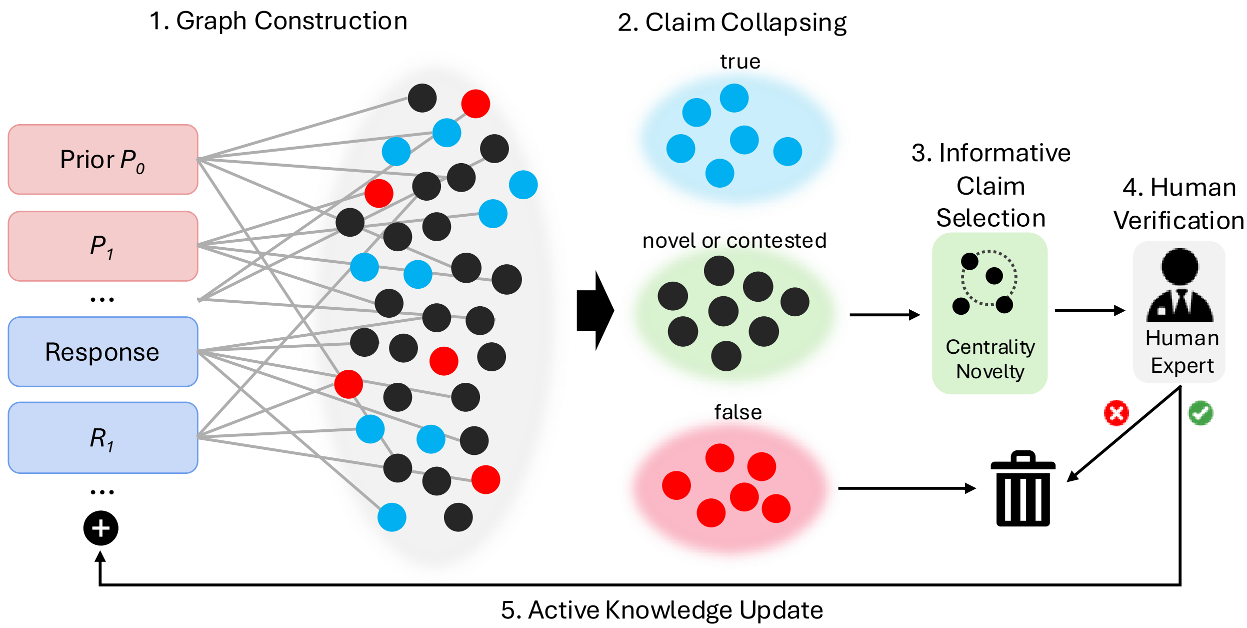

Figure 1: **Overview of our proposed framework.** Priors ($P_0, P_1, \cdots$), i.e., trusted knowledge, are added along with LLM responses ($R_0, R_1, \cdots$), then deconstructed and connected to claims. Edge weights are assigned based on the closeness to the prior. We use weighted graph centrality metrics to "collapse" the claims to clusters: true (high consensus, grounded to priors), novel or contested, and false (consistent, shared hallucinations). Novel and contested claims are considered to be in the LLM's knowledge boundary, and are routed to human experts for validation. Claims judged to be true are added as new priors to iteratively expand the knowledge base.

2 Introduction

Large Language Models (LLMs) have demonstrated rapidly increasing capabilities in reasoning, complex problem solving, and acting as core components in agentic systems. These capabilities have driven widespread adaptation across industries. McKinsey's recent industry analysis [7] reports that more than 78% of organizations now regularly use generative AI (including LLMs) in at least one business function, nearly double the rate from the previous year. Despite this surge in interest, there is prevalent hesitation to deploy LLMs in critical, high-value workflows. While adaptation rate is high, scaled deployment is low; more than 90% of function-specific use cases remain stuck in the pilot stage [1]. The primary bottleneck is reliability. Inaccuracy is constantly cited as the top risk in deploying GenAI systems [4, 8], preventing their use in high-stakes decision making where precision is non-negotiable. LLMs are prone to hallucinations, exhibit high response variance, and are highly sensitive to problem formulation. Furthermore, they present severe security risks; even the most advanced "safety-aligned" models from top labs remain highly vulnerable to sophisticated jailbreak attacks. These combined risks of intrinsic unreliability and extrinsic vulnerability make LLMs unsuitable for core business operations.

The fundamental issue is that LLMs struggle to differentiate between what they genuinely know and what lies on the boundary of their training data. When prompted with out-of-knowledge queries, they often confabulate (hallucinate) rather than abstain ("I don't know the answer to your question"). This creates a complex topology of claims: some LLM-generated claims are "in agreement and factual," while others represent consistent, shared errors (i.e., "in agreement but hallucinated.") Many others are "contested", existing in ambiguous, novel, or low-consensus states.

Current mitigation techniques, such as Retrieval-Augmented Generation (RAG) [3] or uncertainty estimation [6], are insufficient because they still rely heavily on the base model's (unreliable) internal capacity to synthesize and judge information. This necessitates a new layer of external, structured verification. We believe that **grounding LLM outputs against validated external knowledge "priors" combined with human verification** provides the necessary signal for safe and scalable deployment.

We propose a novel graph-based framework (Figure 1) that deconstructs LLM responses into a bipartite graph of claims and sources, measuring both uncertainty and truthfulness through graph centrality analysis. Simply put, a node with high centrality means that it is highly connected and supported by other nodes. Leveraging and extending recent work [2] on graph-based uncertainty estimation, we introduce "knowledge priors"; trusted information sources such as compliance rulebooks, SOPs, or expert opinions. Instead of entirely relying on the LLM's internal knowledge—which is difficult to probe and unreliable—implementing priors can display how much LLM responses are anchored to trusted knowledge. This structure allows us to "cascade" the validation process, tracing claims to group them into "collapsible" clusters based on their connection to trusted priors. The system's core function is to then "collapse" this ambiguity, converting each cluster into a clear determination (true, false, or to be human-verified) by grounding it against the priors. By weighting each edge in the graph based on connectivity (or closeness) to these priors, we move beyond simple consistency checking—which fails to distinguish "in agreement but hallucinated" claims—to true knowledge grounding. Our framework achieves two key objectives:

1. **Knowledge Boundary Identification:** It clearly maps the taxonomy of claims, distinguishing between "grounded" (in agreement and factual), "contradictory" claims (which are collapsed as false), and "novel" claims that are potentially true but unverified.

2. **Iterative Knowledge Expansion:** The system builds and evolves the knowledge boundary by routing only the high-value claims to human experts for verification, turning them into new, known priors. We use a systemic algorithm inspired by active learning [5] to select the most informative claims for human labeling.

Our proposed method creates a dynamically growing, self-correcting knowledge base that balances autonomy with safety. It maximizes the LLM's ability to construct and grow knowledge while minimizing the human effort required for verification. This expansion is optimized using active learning to strategically select the most informative cases for human review, effectively keeping human intervention only where it is absolutely necessary. The remainder of this report details our proposed graph-based framework and its use cases targeted to improving business workflows and procedures.

3 Use Cases

This section highlights some potential use cases of our proposed methods. These use cases leverage our method's ability to autonomously construct and expand knowledge graphs from priors, and delegate crucial validation to humans for safe and reliable knowledge building and decision making. Figure 2 visualizes the graph structures corresponding to these use cases. The use cases identify certain clusters of claims—contradictory claims (Figure 2 left) or novel claims (Figure 2 right)—to delegate to humans for verification.

3.1 Dynamic Compliance and Security Monitoring

Our proposed method can be applied to ensure business operations strictly adhere to internal/external policies and regulations. Priors are defined as the company's compliance rulebooks

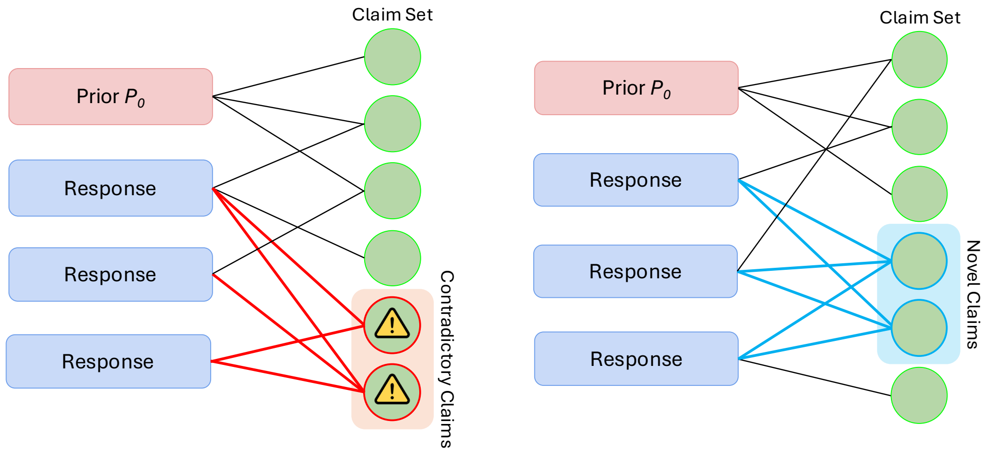

Figure 2: **Knowledge Graph Structures Displaying Claim Clusters.** We illustrate examples of graph structures that show different clusters of claims. (Left) displays clusters of **contradictory claims**, where claims that contradict the prior are also supported by multiple LLM generations. Examples of these would be multiple, repeated cases of safety or compliance violations. (Right) displays a bipartite graph with a cluster of **novel claims**: highly-connected claims that are not supported by priors nor are contradictory. These claims correspond to known-unknowns, or boundary knowledge. Examples of novel claims would be highly requested automation targets or new information frequently brought up in internal communications.

and security protocols. Specialized agents fine-tuned to analyze different data formats monitor employee communications, access logs, and software deployments. These agents turn raw data into natural language claims. Graph construction is conducted with the priors and operational claims from the monitoring agents. Security or compliance violations are detected through "contradictory" claims, such as a file transfer that violates the company's data handling regulation (prior), which are distinguished by their very low centrality score. Depending on the level of violation, such event can be delegated to a security protocol, or routed to a human compliance officer for validation and intervention.

3.2 Identifying Automation Targets in Internal Workflows

Our method can be used to identify targets for automation in task workflows. The prior is a database of all existing automation scripts and APIs in the company's production pipeline. The knowledge bipartite graph is constructed and updated using task requests such as IT tickets. When a new task request arrives, its claims are checked against the priors. For example if a "recalibrate robot arm" automation script exists, the task is grounded and routed to the calibration agent for execution. Conversely, when the system identifies a novel cluster of claims, such as 50 weekly requests for "update inventory for part A", that is not in the prior, the system has identified the "knowledge boundary", a prime automation target which is routed to a human developer to build. Once the new automation pipeline for the task is developed, it is added as a new prior.

3.3 Iterative Expansion of an Internal Knowledge Base

This method is ideal for creating a self-improving, internal knowledge library (e.g., "wiki") that captures and validates internal communications and issues. Here, the prior is the company's existing documentation, such as API documentation, standard procedure manuals, and code readmes. Specialized agents can monitor where knowledge is shared, such as Q&A channels and code pull requests. An example of a claim could be a bug fix request for a codebase. This claim is identified as "novel" since it is an issue not seen in the codebase thus not in the prior. When the system sees this claim has high centrality (e.g. requested > 10 times by multiple individuals), it flags this as uncaptured, high-value information. The claim is then routed to the human (e.g. product owner) for validation. Upon approval, it is iteratively added to the knowledge graph as a prior.

4 Method

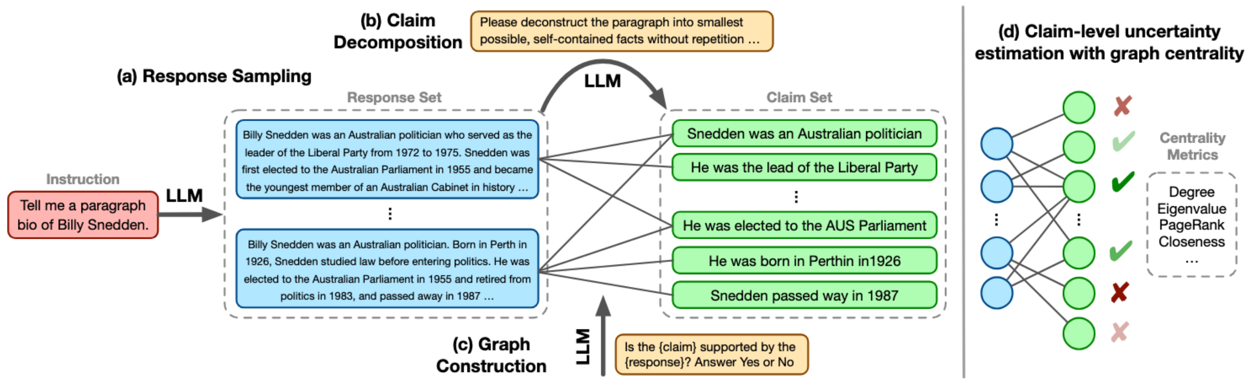

Figure 3: **Overview of the Baseline Graph-Based Uncertainty Method.** (a) Multiple responses are sampled from LLMs and (b) decomposed into sentence-level claims. Then, (c) a bipartite graph is constructed to model the relations between responses and claims. (d) claim-level uncertainty is measured with graph centrality metrics, and a final response is generates with high-scoring claims. Figure from [2].

In this section we detail our proposed weighted bipartite knowledge structure for knowledge boundary identification and human-in-the-loop knowledge expansion. We first discuss the baseline method our model expands on, then present our method and two key functions: knowledge boundary identification and iterative knowledge expansion. A more technical explanation of our method can be found in the appendix.

4.1 Preliminaries

Our work extends the "Graph Uncertainty" frame work introduced by Jiang et al. [2]. Figure 3 illustrates their overall framework. Their method addressed LLM hallucinations by constructing a bipartite graph, a network composed of two types of nodes: **Responses** (sampled outputs from the LLM) and **Claims** (individual facts extracted from the responses). Edges are drawn between a response and a claim if the response supports that claim. The key insight is that true claims tend to be more "central" in the graph, supported by many different responses. By calculating graph centrality scores, i.e., how well-connected a node is, their framework can estimate the uncertainty of each claim. High centrality scores indicate a likely true claim, while low centrality suggests a potential hallucination.

While effective at filtering unsupported claims, this baseline relies entirely on the LLM's own internal capacity, or consistency. It cannot distinguish between a fact and a "shared hallucination": a false claim that is consistently generated by the LLM. If an LLM confidently and repeatedly makes the same mistake, that false claim will receive a high centrality score. Thus, consistency alone is not a guarantee of factuality, especially for knowledge outside the model's knowledge boundary.

4.2 Grounded Graph Construction

To overcome this limitation, we introduce **"knowledge priors"**: trusted external information sources such as rulebooks, Standard Operating Procedures (SOPs), or expert-verified documents. We integrate these priors directly into the graph structure to serve as "anchors" of truth.

Unlike the baseline method, which treats all connections equally, our method constructs a **weighted bipartite graph**. We assign different weights to edges based on their trusted source:

1. **Anchor Links** (Highest Weight): Connections originating directly from the prior. These are treated as "ground truth".
2. **Agreement Links** (High Weight): Connections where an LLM response correctly identifies a claim that is also present in the prior.
3. **Neutral Links** (Standard Weight): Connections between LLM responses and "novel" claims that are not present in the prior.
4. **Contradictory Links** (Low Weight): Connections to claims that contradict the prior.

The weights are integrated into the graph centrality metrics to produce weighted centrality scores. Refer to the appendix for detailed explanations. By using a weighted centrality metric, claims that are closely connected to the priors receive significantly higher scores. This ensures that highly ranked claims are not just consistent, but actively grounded in verified knowledge, i.e., priors.

4.3 Knowledge Boundary Identification

Constructing a weighted graph structure enables the identification of claims at different positions in the LLM's knowledge space. Highly scored claims reside at the known space. Contradictory claims, which receive very low scores, are hallucinations and can be discarded from the graph. Crucially, we can identify "boundary" claims; those with intermediate scores between the known claims and contradictory claims. These claims are not well supported by priors and sparsely connected to other nodes in the graph. We consider these claims as in the knowledge boundary, the "known-unknowns". These can be true claims that are outside the knowledge and the priors, or just hallucinations. Further verification is required to judge which category these belong to.

4.4 Iterative Knowledge Expansion

Instead of discarding these boundary claims identified from the previous step, our system routes them to human experts for validation. A human can determine whether to boost or eliminate these claims. Verified claims can also be added to the graph as a new prior. Separately, new priors can be added simply when new information is introduced. Edge weights are re-calculated after each update. This process creates a scalable, self-improving cycle where the trusted knowledge base continuously expands with minimal, targeted human effort.

Informative Claim Selection. To maximize the value of human intervention and minimize the cost of human effort, we use an intelligent selection algorithm to prioritize only the most informative claims for review. This is inspired by works in active learning [5], where the most informative data are selected for human labeling in an incremental learning process. This system acts like a smart filter, focusing on two high-value targets. First, it finds "source" claims that are central to a large cluster of other unverified claims, allowing a single human verification to collapse and resolve an entire topic. Second, it explores new frontiers by selecting truly "novel" claims, which is how the system expands its knowledge base by discovering and verifying entirely new information. Technical details of the selection algorithm can be found in the appendix.

4.5 Multi-Agent Integration

A key extension of our framework is multi-agent integration. This approach replaces the standard response sampling from a single LLM with a more diverse set of inputs from multiple, specialized agents. Instead of generating potentially redundant claims from one model, this method leverages agents with different, specialized knowledge. Because each agent possesses distinct expertise, claims that are supported by several of these specialized agents is considered significantly more powerful and reliable. This inter-agent agreement provides a stronger signal of a claim's validity than consistency from a single source.

A Method - Technical Details

This section details the formal construction of our weighted bipartite graph and the specific centrality metrics used for grounded uncertainty estimation.

Graph Construction and Notation. Following Jiang et al., [2] we construct a bipartite graph $G = ((\mathcal{R} \cup \mathcal{P}, \mathcal{C}), \mathcal{E})$ where $\mathcal{R}$ is the set of sampled LLM responses, $\mathcal{P} = \{P_0, P_1, \cdots\}$ is the set of trusted priors, and $\mathcal{C}$ is the set of unique atomic claims decomposed from both $\mathcal{R}$ and $\mathcal{P}$. An edge $e = (u, v) \in E$ exists if a source node $u \in (\mathcal{R} \cup \mathcal{P})$ entails a claim node $v \in \mathcal{C}$.

Weighted Adjacency via Priors. Unlike the original unweighted approach, we define an edge weighting function $W : E \to \mathbb{R}^+$ to encode closeness to priors. Let $c^* \in \mathcal{C}_P$ denote a claim that is entailed by a prior P_i. We assign edge weights w_{uv} as follows:

$$w_{uv} = \begin{cases} w_{anchor} & \text{if } u \in \mathcal{P} \text{ and } v \in \mathcal{C}_P \text{ (Anchor Link)} \\ w_{agree} & \text{if } u \in \mathcal{R} \text{ and } v \in \mathcal{C}_P \text{ (Agreement Link)} \\ w_{neutral} & \text{if } u \in \mathcal{R} \text{ and } v \notin \mathcal{C}_P \text{ (Neutral Link)} \\ w_{contra} & \text{if } v \text{ contradicts } \mathcal{P} \text{ (Contradictory Link)} \end{cases}$$

where $w_{anchor} > w_{agree} > w_{neutral} > w_{contra}$ (e.g., 3.0, 2.0, 1.0, 0.5). This ensures that paths through priors have the lowest traversal cost, defined as $cost(u, v) = 1/w_{uv}$.

Grounded Closeness Centrality. We adopt Closeness Centrality (C_C) as our primary uncertainty metric, following the findings of Jiang et al. [2] that it is highly effective at detecting false claims. To account for potential graph disconnectedness, we utilize the modified closeness centrality variant, adapted for our weighted structure:

$$C_C(v) = \frac{N - 1}{\sum_{u \in V} d_W(v, u)} \cdot \frac{|V_v|}{N}$$

Here, N is the total number of nodes, $|V_v|$ is the size of the connected component containing v, and crucially, $d_W(v, u)$ is the shortest path distance between nodes v and u calculated using the edge costs $1/w_{uv}$.

By using Dijkstra's algorithm to compute d_W, this metric naturally assigns higher centrality to claims v that are reachable via low-cost "Anchor" and "Agreement" links. Claims that mainly connect via high-cost "Neutral" or "Contradictory" links will have larger average distances and thus lower C_C scores. This effectively pushes out ungrounded hallucinations while boosting grounded facts and high-consensus novel claims for established knowledge and expansion candidates, respectively.

Informative Claim Selection Algorithm. To optimize the Iterative Knowledge Expansion (Section 4.4), we prioritize "boundary" claims for human review using a hybrid active learning strategy. This algorithm selects claims that are most informative, defined as claims that resolve the most ambiguity, or expand the knowledge boundary into novel areas.

We first define the set of candidate claims $C_{contested}$ for selection, as all claims are not entailed by a prior:

$$\mathcal{C}_{contested} = \mathcal{C} \setminus \mathcal{C}_P$$

Note that $\mathcal{C}_{contested}$ can be selected by excluding $\mathcal{C}_P$, or simply selecting claims below a certain centrality score δ as such: $\mathcal{C}_{contested} = (C_C < \delta) \forall \mathcal{C}$.

For each claim $c \in \mathcal{C}_{contested}$, we compute a $PriorityScore(c)$ as a weighted sum of two normalized scores:

$$PriorityScore(c) = \lambda_1 \cdot S_{Centrality}(c) + \lambda_2 \cdot S_{Novelty}(c)$$

Where λ_i are tunable hyperparameters to balance the selection strategy. Note that these scores can be separated to individually sample for different objectives.

1. **Centrality Score ($S_{centrality}$):** This score prioritizes "sources" inside the unverified part of the graph, aiming to collapse the clusters of related unverified claims. Give a subgraph $G_{contested} = ((\mathcal{R}, C_{contested}), \mathcal{E}')$ consisting only of LLM responses and contested claims, we compute the Betweenness Centrality $C_B(c, G_{contested})$ for c in this subgraph:

$$S_{centrality}(c) = \frac{C_B(c, G_{contested})}{\max_{c' \in C_{contested}} C_B(c', G_{contested})}$$

2. **Novelty Score ($S_{novelty}$):** This score identifies novel claims that are most distant from any known knowledge. We use the shortest path distance metric ($d_W(u, v)$) to find the distance from a claim c to the nearest true claim c^*:

$$d_{min}(c) = \min_{c^* \in \mathcal{C}_P} d_W(c, c^*)$$

The score is normalized by the maximum observed distance:

$$S_{novelty}(c) = \frac{d_{min}(c)}{\max_{c' \in C_{contested}} d_{min}(c')}$$

Using this scoring metric, we route the top-k claims to the human expert for verification. This ensures that human effort is directed at the most critical and novel claims, maximizing the efficiency of the knowledge expansion loop.

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